

Google Cloud Platform Fundamentals

Lab Guide for Anthos

Run the below commands in the macOS Terminal

```
export PROJECT_ID=<PROJECT ID> # Replace this with your GCP project ID
export REGION=<REGION ID> # Replace this with a valid GCP region
```

```
gcloud config set project $PROJECT_ID
gcloud config set compute/region $REGION
```

Enable APIs

```
gcloud services enable \
  container.googleapis.com \
  gkeconnect.googleapis.com \
  gkehub.googleapis.com \
  cloudresourcemanager.googleapis.com
```

Launch GKE Cluster

```
gcloud container clusters create cloud-cluster \
  --machine-type=n1-standard-1 \
  --num-nodes=1
```

Launch Minikube. Refer to the docs at <https://minikube.sigs.k8s.io/docs/>
minikube start

Create GCP Service Account

```
gcloud iam service-accounts create anthos-hub
```

Google Cloud Platform Fundamentals

Add IAM Role to Service Account

```
gcloud projects add-iam-policy-binding $PROJECT_ID \  
  --member="serviceAccount:anthos-hub@$PROJECT_ID.iam.gserviceaccount.com" \  
  --role="roles/gkehub.connect"
```

Download the Service Account JSON Key

```
gcloud iam service-accounts keys create "./anthos-hub-svc.json" \  
  --iam-account="anthos-hub@$PROJECT_ID.iam.gserviceaccount.com" \  
  --project=$PROJECT_ID
```

Register cluster with Anthos

```
URI=` gcloud container clusters list --filter='name=cloud-cluster' --uri`
```

```
gcloud container hub memberships register cloud-cluster \  
  --gke-uri=$URI \  
  --service-account-key-file=./anthos-hub-svc.json
```

List Membership

```
gcloud container hub memberships list
```

Register Minikube with Anthos

```
gcloud container hub memberships register local-cluster \  
  --service-account-key-file=./anthos-hub-svc.json \  
  --kubeconfig=~/.kube/config \  
  --context=minikube
```

List Membership

```
gcloud container hub memberships list
```



Google Cloud Platform Fundamentals

Create Kubernetes Role

```
kubectl config use-context minikube

cat <<EOF > cloud-console-reader.yaml
kind: ClusterRole
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: cloud-console-reader
rules:
- apiGroups: [""]
  resources: ["nodes", "persistentvolumes"]
  verbs: ["get", "list", "watch"]
- apiGroups: ["storage.k8s.io"]
  resources: ["storageclasses"]
  verbs: ["get", "list", "watch"]
EOF
kubectl apply -f cloud-console-reader.yaml
```

Create RoleBinding

```
kubectl create serviceaccount local-cluster

kubectl create clusterrolebinding local-cluster-anthos-view \
  --clusterrole view \
  --serviceaccount default:local-cluster

kubectl create clusterrolebinding cloud-console-reader-binding \
  --clusterrole cloud-console-reader \
  --serviceaccount default:local-cluster
```



Google Cloud Platform Fundamentals

Get the Token

```
SECRET_NAME=$(kubectl get serviceaccount local-cluster -o jsonpath='{$.secrets[0].name}')
```

Copy the secret and paste it in the console

```
kubectl get secret ${SECRET_NAME} -o jsonpath='{$.data.token}' | base64 --decode
```

Delete Membership

```
gcloud container hub memberships delete cloud-cluster  
gcloud container hub memberships delete local-cluster
```

Clean up

```
gcloud container clusters delete cloud-cluster --project=${PROJECT_ID}  
gcloud iam service-accounts delete anthos-hub@${PROJECT_ID}.iam.gserviceaccount.com  
minikube delete
```